Tharindu Disnaka Gamage

Mechatronics System Engineering Undergraduate

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ABOUT ME

I am a Mechatronics Systems Engineering undergraduate at the University of Moratuwa, specializing in robotics, automation, and control systems. Passionate about solving complex problems through innovative solutions and teamwork.

Portfolio Link - https://tharindu-disnaka-portfolio.vercel.app/

EDUCATION

2022 - 2026

UNIVERSITY OF MORATUWA

BSc Eng (Hons) Mechanical Engineering specialized in Mechatronics System Engineering

• CGPA = 3.27

2017 - 2020

GAMINI CENTRAL COLLEGE, NUWARA ELIYA

Physical Sciences

• Achieved 3A's in Physical Science Stream (Mathematics, Physics, Chemistry)

• Z-Score: 2.1021

2005 - 2016

OUR LADY'S UPPER SCHOOL

Ordinary Level (O-Level)

• 9A's - GCE Ordinary Level (O-level)

• Leader Seaman in National Cadet Corps (NCC) Sri Lanka

• Emergency First Aider trained by the Sri Lanka Red Cross

WORK EXPERIENCE

2024 December to 2025 June

6 MONTHS TRAINING - COLOMBO INTERNATIONAL CONTAINER TERMINALS

During my internship at Colombo International Container Terminals (CICT), I gained hands-on experience in industrial automation, mechanical design, and maintenance systems within a large-scale port environment. I worked on real-world engineering projects including PLC-based automation, machine design, and vision-based systems, contributing to improvements in operational efficiency and reliability.

PROJECTS

Semester 5, 2024 - 2025

DIYAKAWA 2.0 - AUTONOMOUS UNDERWATER ROBOT:

Designed and improved an Autonomous Underwater Vehicle (AUV) for the Singapore AUV Challenge 2025. Contributed to structural design and system integration. Achieved 10th place out of 48 international teams.

Semester 4, 2024 - 2025

DIYAKAWA - UNDERWATER AUTONOMOUS ROBOT:

Designed and built an Autonomous Underwater Vehicle (AUV) for the Singapore AUV Challenge 2024. Worked as the Structural Design Engineer, responsible for creating the external body using SolidWorks and Fusion 360. The team placed 11th out of 100+ teams worldwide.

Semester 4, 2024

REVERSE ENGINEERING OF A BICYCLE DISK BRAKE SYSTEM:

Dissected and optimized a bicycle disc brake system. Conducted material tests such as density and Rockwell hardness testing, and created 3D models in Autodesk Fusion 360. Optimizations improved the safety factor and reduced weight by 8.3 percent.

Semester 5, 2024

QUADRUPED ROBOT DOG - MECHATRONICS SYSTEM DESIGN PROJECT:

Designed, developed, and implemented a 12-DOF quadruped robot capable of walking and trotting using inverse kinematics and gait planning. Integrated ESP32-based control, Matlab/Simulink simulations, and 3D-printed mechanical design.

Semester 6, Internship

OIL DISTRIBUTION AUTOMATION SYSTEM ENHANCEMENT (CICT INTERNSHIP):

Improved an industrial oil dispensing automation system using Delta PLC, HMI, load cells, and barcode integration. Calibrated sensors, resolved control logic issues, and implemented anti-theft flow detection. Enhanced system reliability and data handling, improving operational accuracy.

Semester 6, Internship

TRUCK ALIGNMENT DETECTION AND FEEDBACK SYSTEM (CICT INTERNSHIP):

Developed a vision-based truck alignment system using Jetson and YOLO for object detection. Implemented real-time data communication with Arduino via Ethernet and designed a feedback display system to guide drivers accurately.

Semester 6, Internship

AUTOMATED ACB RE-POWERING SYSTEM (PLC-BASED) (CICT INTERNSHIP):

Contributed to the automation of ACB re-powering in a DC 680V distribution system using a Siemens LOGO PLC. Modified control logic, installed hardware, and validated system operation, reducing manual recovery time and improving port operational efficiency.

Semester 7 & 8, 2025 - 2026

AUTONOMOUS ASSISTIVE ROBOT FOR ELDERLY CARE (FINAL YEAR PROJECT):

Designed and developed an autonomous assistive robotic system for safe transfer and continuous health Monitoring of elderly individuals with sarcopenia. Integrated mobile navigation, mechanical lifting mechanism, and IoT-based health monitoring with ROS2 simulation and human tracking.

VOLUNTEERING

February 2024 - Present

PROGRAM AND LOGISTICS COMMITTEE LEADER

BOT TALK 2.0 EVENT, IEEE RAS (Robotics and Automation Society) Student Branch Chapter of University of Moratuwa

November 2023 - Present

VOLUNTEER MANAGEMENT COMMITTEE MEMBER

IEEE RAS (Robotics and Automation Society) Student Branch Chapter of University of Moratuwa

2023 - Present

IESL STUDENT MEMBER

Project Demonstrator Techno Sri Lanka 2023

SKILLS

- Programming: Python, C/C++, MATLAB, PLC ladder
- Embedded Systems: Arduino, ESP32, STM32, Jetson (NVIDIA), Microcontroller Programming
- PLC & Automation: Siemens, Delta PLC, HMI Design, Industrial Automation Systems
- Robotics & Control: ROS2, Inverse Kinematics, Gait Planning, Control Systems
- Computer Vision: Yolo, Image Processing, Object Detection
- CAD & Simulation: SolidWorks, Autodesk Fusion 360, MATLAB/Simulink
- Mechanism Design, Structural Design, FEA (Stress Analysis).

REFERENCES

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